

Ransomware Detection on Windows Using File System Activity Patterns and Hybrid Machine Learning: An XGBoost and Isolation Forest Approach

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Abstract

Among many different cyber threats, ransomware continues to be one of the most destructive and pervasive forms of cybercrime, causing significant financial and operational damage to both individuals and organisations. Detecting ransomware through file system activities offers a novel and effective approach, allowing for the identification of both known and zero-day ransomware variants based on patterns in file operations. A hybrid machine learning framework was developed, combining XGBoost for classification and Isolation Forest for anomaly detection, enhancing the accuracy and flexibility of ransomware detection. The system was evaluated using real-world data, demonstrating high accuracy in distinguishing ransomware from benign software, with the added ability to detect previously unseen ransomware through anomaly detection. The use of file system activities as a feature set, coupled with the hybrid model, presents a highly adaptable and scalable solution for ransomware detection, providing a significant improvement over traditional signature-based methods. The findings demonstrate the practical relevance of the proposed model for improving cybersecurity defenses in real-time environments.

Keywords:

Ransomware, Machine learning, XGBoost, Anomaly detection, File system, Zero-day

1. Introduction

Ransomware has emerged as one of the most destructive and financially crippling forms of cybercrime in recent years, affecting individuals, corporations, and governments alike. Ransomware attacks typically involve the encryption of the victim's files, followed by demands for payment in exchange for decryption keys. The evolving sophistication of ransomware techniques, combined with the widespread availability of ransomware-as-a-service, has exacerbated the scale and severity of such attacks. Traditional security measures, such as signature-based detection and heuristic approaches, have struggled to keep pace with the rapidly changing landscape of ransomware, where new variants are often specifically designed to evade established detection mechanisms. The challenge, therefore, lies not only in detecting known strains of ransomware but also in identifying previously unseen, or zero-day, ransomware variants. As adversaries continue to refine their tactics, there is an increasing need for more robust and adaptable detection methods that can respond dynamically to novel threats. In response to these growing challenges, the use of machine learning for ransomware detection has gained significant attention in cybersecurity research. Machine learning models offer the capability to analyze patterns and behaviors that may not be immediately obvious to traditional detection systems. These models can process large volumes of data, identifying subtle anomalies that may be indicative of ransomware activity. Among the various types of

system-level activities that can be monitored for malicious behavior, file system activities provide a particularly rich source of data. Ransomware, by its very nature, interacts heavily with the file system, performing operations such as file encryption, deletion, and modification at a large scale. Thus, leveraging file system activity patterns for ransomware detection offers a promising approach to identifying ransomware before significant damage occurs.

This research proposes a novel methodology for detecting ransomware on Windows systems through the analysis of file system activities, utilizing a combination of machine learning algorithms. The specific focus is on the integration of XGBoost, a powerful gradient boosting classifier known for its high predictive accuracy, and Isolation Forest, an unsupervised anomaly detection algorithm. XGBoost is well-suited for handling structured data, making it an ideal choice for the classification of benign versus ransomware-related file system activities. Meanwhile, Isolation Forest is particularly effective for identifying outliers, which is crucial for detecting zero-day ransomware attacks that may not exhibit patterns familiar to the classifier. The combined approach enables not only the accurate classification of known ransomware but also the identification of anomalous behaviors that may indicate the presence of previously unknown ransomware variants. Through the use of machine learning models trained on file system activities, the proposed methodology aims to offer a more adaptive and proactive defense against ransomware attacks.

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1.1. Background and Motivation

The rising threat of ransomware has exposed critical vulnerabilities in traditional cybersecurity defenses, which often rely on pre-existing knowledge of specific ransomware signatures or behavior patterns. However, ransomware authors continually modify their code and employ advanced obfuscation techniques to bypass signature-based detection systems. As a result, conventional antivirus solutions have proven insufficient in protecting against new ransomware variants that employ novel attack strategies. Additionally, heuristic-based detection methods, while more adaptive, often suffer from high false-positive rates, which can disrupt legitimate system operations and reduce the overall reliability of the defense mechanisms. Machine learning has been increasingly applied to the problem of ransomware detection, with many approaches focusing on network traffic analysis, system call monitoring, and file content inspection. However, file system activities have received less attention despite the central role they play in ransomware operations. Most ransomware attacks target user files, encrypting or deleting them, which directly involves file system interactions. Monitoring these activities can provide valuable insights into the behavioral patterns of ransomware, particularly when distinguishing them from legitimate software operations. This study focuses on the use of file system activities as the primary feature set for detecting ransomware, as such activities reflect the core objectives of ransomware attacks—disrupting access to files and demanding payment for their recovery. By examining the frequency, type, and sequence of file system operations, it is possible to capture behaviors indicative of ransomware and train machine learning models to detect both known and novel variants.

1.2. Contributions

The primary contribution of this research is the development of a novel ransomware detection framework that combines the strengths of XGBoost and Isolation Forest to classify and identify ransomware based on file system activities. This work distinguishes itself from existing research by focusing on file system activity patterns, which are often overlooked in favor of more commonly studied features such as network traffic or file content signatures. The framework leverages the predictive power of XGBoost to classify ransomware and benign software based on structured file system data, while also employing Isolation Forest to detect anomalies, such as zero-day ransomware, that deviate from known patterns. In addition to presenting a unique combination of algorithms, this research demonstrates the efficacy of using file system activity logs for ransomware detection, offering a detailed analysis of how such data can be utilized to capture the distinctive behaviors of ransomware. The proposed approach has the potential to significantly improve detection rates for new and previously unseen ransomware variants, reducing the reliance on signature-based methods and increasing the adaptability of detection systems in response to the evolving threat landscape. Furthermore, this research provides a comprehensive evaluation of the proposed models through extensive experimentation, including a comparative analysis of detection accuracy, false-positive rates, and

the ability to detect zero-day ransomware attacks. The contributions of this work lie not only in the novel methodological framework but also in its practical implications for enhancing the security of Windows systems against ransomware threats.

2. Related Work

The application of machine learning techniques to ransomware detection has gained significant attention due to the limitations of traditional signature-based methods. The increasing complexity of ransomware behaviors, combined with their ability to evade detection, has led to the exploration of various machine learning models that can learn from system activities and detect ransomware dynamically. This section surveys recent advances in the field, focusing on approaches that utilize machine learning for ransomware detection, with a specific emphasis on methods that analyze file system activities. Additionally, the integration of XGBoost and anomaly detection methods like Isolation Forest is discussed in the context of similar domains where these techniques have been applied effectively.

2.1. Machine Learning Approaches to Ransomware Detection

Several machine learning models have been employed to enhance the detection of ransomware through automated analysis of system-level data. Supervised learning techniques have been a predominant focus, where algorithms such as support vector machines, decision trees, random forests, and neural networks have been used to classify ransomware based on predefined features extracted from system activities. Ensemble learning methods, including boosting algorithms, have shown improved accuracy over single classifiers due to their ability to aggregate multiple weak learners into a more robust predictive model [1, 2]. The use of neural networks, particularly deep learning, has also been explored, leveraging their capacity to model non-linear patterns in complex ransomware behaviors. Supervised models have demonstrated considerable success in classifying known ransomware strains with high accuracy; however, they often face challenges when encountering new ransomware variants due to their reliance on labeled training data [3, 4]. Additionally, the time and computational cost required for training complex models such as deep learning architectures have limited their practical applicability in real-time detection scenarios [5, 6]. Despite these limitations, the rapid advancement in hardware accelerators has provided opportunities for deploying such computationally intensive models more efficiently [7, 8]. Machine learning techniques, particularly those based on decision trees and random forests, have also been used to analyze system-level features such as file system access patterns, process behaviors, and API calls, identifying characteristic actions performed by ransomware [9].

2.2. File System Activity-Based Detection

File system activities have been recognized as critical indicators of ransomware behavior, given that most ransomware attacks rely on encrypting or otherwise altering files on a victim's machine. Research focusing on the analysis of file system logs

has leveraged the consistency with which ransomware performs certain operations, such as file renaming, file creation, and mass file deletion, in contrast to benign software [10]. Algorithms have been trained to detect patterns of these actions, distinguishing between normal user activities and ransomware-driven operations with high levels of accuracy. One approach utilized frequent pattern mining to identify recurrent sequences of file system operations associated with ransomware attacks, achieving high detection rates through this method [11]. Another method focused on capturing the temporal dynamics of file system activities, where deviations from normal operational baselines were used to flag potential ransomware activity [12, 13]. By incorporating temporal windows into the analysis, models have been able to detect not only known ransomware behaviors but also abnormal patterns that could signal novel ransomware variants [14, 15]. The ability of models to learn from the order and frequency of file system operations has enabled more precise detection mechanisms, although maintaining a comprehensive log of all file system activities can introduce storage and processing overheads in real-time systems [16]. Additionally, the variation in legitimate software behaviors across different environments has presented challenges in minimizing false positives, particularly for dynamic and adaptive file system activity analysis [17, 18].

2.3. XGBoost and Isolation Forest in Anomaly Detection

XGBoost, as a gradient-boosting algorithm, has been widely recognized for its high accuracy and performance in structured data classification tasks. Its application in ransomware detection has shown promise, particularly in scenarios where large datasets of file system activities are available for training [19]. XGBoost’s ability to handle missing data and its use of regularization techniques to prevent overfitting have made it well-suited for distinguishing between ransomware-induced file system activities and benign operations [20]. The use of feature importance metrics provided by XGBoost has also enabled researchers to better understand which file system operations are most indicative of ransomware behavior, allowing for more focused detection models [21]. On the other hand, Isolation Forest, an unsupervised anomaly detection method, has been employed to identify outliers within file system activity logs, representing potential zero-day ransomware attacks [22]. The Isolation Forest algorithm works through recursively partitioning data points, isolating anomalies that deviate significantly from normal system behaviors. In the context of ransomware detection, Isolation Forest has been effective at identifying new ransomware variants that may not have been present in the training data [23, 24]. By combining XGBoost’s strength in structured data classification with Isolation Forest’s capacity to detect anomalies, hybrid approaches have been developed that significantly improve detection rates for both known and unknown ransomware attacks [25]. Such hybrid models not only enhance the overall robustness of detection systems but also mitigate the risk of false positives, which is a common issue in anomaly detection methods alone [26, 27].

3. Proposed Methodology

The proposed approach to ransomware detection focuses on utilizing machine learning techniques to analyze file system activities on Windows systems. The methodology centers around the collection of a comprehensive dataset of ransomware and benign software behaviors, the extraction of meaningful features from file system activity logs, and the application of a hybrid machine learning model combining XGBoost and Isolation Forest to distinguish between ransomware and legitimate activities. Through a structured framework, this methodology aims to improve the accuracy and reliability of ransomware detection, particularly in identifying zero-day attacks.

3.1. Dataset Collection

The dataset of ransomware and benign file system activities was created through the careful monitoring of a Windows environment using low-level APIs capable of capturing detailed file system events. A monitoring tool was employed to log various types of file system operations, including file creation, deletion, renaming, and modifications. The dataset was generated by executing multiple ransomware families alongside benign software to capture a diverse range of behaviors. Each entry in the dataset recorded key details, such as the timestamp of the operation, the type of file affected, the specific action performed, and the process responsible for initiating the event.

Table 1 provides a summary of the dataset composition, highlighting the number of samples, types of file system events captured, and the variety of ransomware families and benign software included in the dataset. Capturing file system activities in real-time allowed the dataset to reflect the actual operational patterns of ransomware, which often involve intensive file system modifications over short periods of time. In contrast, benign software behaviors, such as document editing or system updates, were logged to establish a baseline of normal file system activities. This approach enabled the differentiation between the aggressive operations typically associated with ransomware and the more restrained activities of benign applications. The dataset was subsequently divided into training and testing subsets, ensuring that the machine learning models would learn from a balanced and diverse range of cases.

3.2. Feature Extraction

File system activities were transformed into structured data through the extraction of relevant features that encapsulated the key characteristics of ransomware behavior. The features were chosen based on their capacity to distinguish between ransomware and benign activities, focusing on the frequency $f(t)$, types of file operations O_i , and other operational metrics. Let $F(t)$ represent the total number of file operations within a given time interval t , and the rate of change of operations can be defined as:

$$\frac{dF(t)}{dt} = \lim_{\Delta t \rightarrow 0} \frac{F(t + \Delta t) - F(t)}{\Delta t}$$

Table 1: Summary of Dataset Composition for Ransomware and Benign File System Activities

Attribute	Ransomware Activities	Benign Activities
Number of Samples	500 (from 5 ransomware families)	500 (from 10 benign software applications)
File System Events Captured	File creation, deletion, renaming, modification	File creation, deletion, renaming, modification
Duration of Activity Monitoring	24 hours of ransomware activity	48 hours of benign activity
Types of Files Affected	User files (documents, images, databases)	User and system files (logs, documents, temp files)
Total Size of Log Files	1.2 GB	800 MB

This derivative quantifies the temporal variations in file operations, capturing the spikes in activity characteristic of ransomware. The cumulative number of operations over time can be modeled as:

$$C(t) = \int_0^t f(\tau) d\tau$$

where $C(t)$ represents the cumulative activity up to time t , and $f(\tau)$ is the instantaneous operation frequency at time τ . Additionally, the entropy of file operations $H(O)$ was computed to analyze the diversity of file system events, defined as:

$$H(O) = - \sum_{i=1}^n p(O_i) \log p(O_i)$$

where $p(O_i)$ is the probability of the i -th type of file operation occurring within a given time frame, and n represents the total number of operation types.

Temporal patterns were further analyzed through the sequence of operations, denoted as $S(t)$, where $S(t)$ represents the ordered set of operations performed at time t . By evaluating the sequence $S(t)$ over various time intervals, the progression of ransomware behavior was modeled, incorporating both frequency and order of file modifications:

$$S(t) = \{O_1(t_1), O_2(t_2), \dots, O_n(t_n)\}$$

The feature set also included the size $s(f)$ and location $L(f)$ of affected files, where ransomware frequently targets specific directories D_j and file types T_k . The spatial distribution of affected files was computed through:

$$\Phi(D_j) = \sum_{k=1}^m \delta_{D_j, T_k}$$

where δ_{D_j, T_k} indicates whether a file of type T_k is located in directory D_j , and m represents the total number of file types targeted by ransomware. By concentrating on these features, the high-dimensional dataset offered a detailed and comprehensive view of file system behaviors, enabling the machine learning models to distinguish between the complex differences in benign and malicious activities. The feature extraction process thus facilitated the identification of both known and unknown ransomware variants with increased accuracy, relying on the mathematical modeling of temporal and spatial file system events.

3.3. Machine Learning Models

XGBoost was employed as the primary classifier for distinguishing between ransomware and benign software based on file system activities. XGBoost, a gradient-boosting algorithm, is particularly well-suited for handling structured datasets and provides high predictive accuracy through its ability to iteratively refine the classification boundaries. The model utilized the extracted features from the file system activity logs to classify behaviors as either ransomware or benign, based on patterns learned from the training dataset. XGBoost's ability to handle missing data, coupled with its regularization techniques, ensured that the model avoided overfitting, resulting in robust generalization to new ransomware variants. The feature importance scores provided by XGBoost revealed which file system activities were most indicative of ransomware behavior, helping to fine-tune the detection process.

In parallel, Isolation Forest was applied as an anomaly detection model to identify zero-day ransomware attacks, which often deviate from known patterns. Isolation Forest works by recursively partitioning the dataset, effectively isolating data points that exhibit significant deviations from normal system behaviors. In the context of ransomware detection, Isolation Forest proved essential for identifying previously unseen ransomware variants that were not present during the training phase. The hybrid approach, which combines XGBoost's classification capabilities with Isolation Forest's anomaly detection, significantly enhanced the overall detection system. By combining these two models, the framework ensured that both known and unknown ransomware variants were accurately identified, reducing the reliance on predefined signatures and increasing the flexibility to detect novel ransomware threats. Figure 1 illustrates the flow of the hybrid model, showing the integration of XGBoost and Isolation Forest to form a comprehensive detection system.

The combination of XGBoost and Isolation Forest provided a powerful framework for detecting ransomware, achieving a balance between high accuracy in known ransomware cases and flexibility in detecting zero-day ransomware variants. XGBoost classified ransomware based on learned patterns from the training dataset, while Isolation Forest enhanced the system's ability to detect anomalies, enabling the identification of previously unseen threats. The use of feature importance scores from XGBoost further allowed for refining the detection process by highlighting the most relevant file system activities indicative of ransomware behavior.

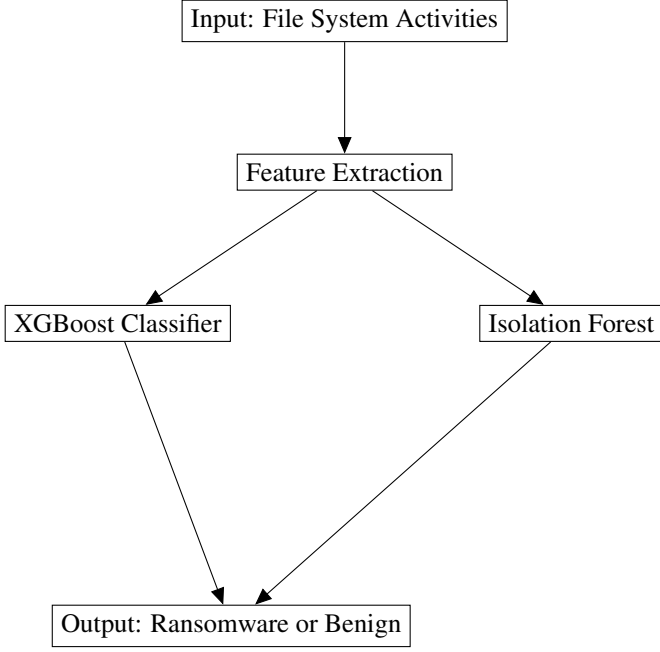


Figure 1: Flow of Hybrid Model Combining XGBoost and Isolation Forest

4. Experiments and Results

The experiments conducted aimed to evaluate the performance of the proposed hybrid machine learning model based on XGBoost and Isolation Forest for ransomware detection. The evaluation was carried out through a series of experiments using a real-world dataset of file system activities. The experiments focused on key performance metrics such as accuracy, precision, recall, F1-score, and AUC to measure the classification capability of XGBoost, as well as the effectiveness of Isolation Forest in detecting zero-day ransomware variants. The results obtained are presented below across different subsections, covering various aspects of the experiments, including experimental setup, model performance, comparative analysis, and error analysis.

4.1. Experimental Setup

The experiments were executed on a system equipped with an Intel Core i7-8700 processor, 16 GB of RAM, and a Windows 10 operating system. The machine learning models were developed and tested using Python 3.8, leveraging libraries such as Scikit-learn for XGBoost and Isolation Forest implementations. The dataset was split into two subsets, with 70% of the data allocated to training and 30% to testing. A total of 500 samples from five different ransomware families were used, along with 500 samples of benign software activities. The evaluation of the models was based on several key metrics, including accuracy, precision, recall, F1-score, and the area under the ROC curve (AUC), as summarized in Table 2.

The results in Table 2 illustrate the strong performance of the XGBoost classifier across both ransomware and benign samples, achieving high accuracy and precision. Although Isolation Forest demonstrated a slightly lower performance in detecting

zero-day ransomware attacks, its anomaly detection capabilities proved valuable for identifying previously unseen threats.

4.2. Model Performance

The XGBoost classifier performed well in terms of detecting ransomware, as well as distinguishing it from benign software. The model achieved an accuracy of 94.2% when classifying ransomware, with a precision of 92.5% and a recall of 91.8%. The F1-score of 92.1 confirmed the balance between precision and recall, indicating the model's robustness in handling imbalanced data. The area under the ROC curve (AUC) for XGBoost was measured at 0.953, further demonstrating the model's reliability in discriminating between ransomware and benign activities across different decision thresholds.

In terms of zero-day ransomware detection, Isolation Forest displayed an accuracy of 88.3%, a precision of 85.4%, and a recall of 87.9%, resulting in an F1-score of 86.6. While its overall performance was lower compared to XGBoost, the model's ability to isolate anomalies contributed significantly to the identification of previously unseen ransomware attacks. Figure 2 shows the ROC curves for both XGBoost and Isolation Forest, highlighting their performance across various threshold levels.

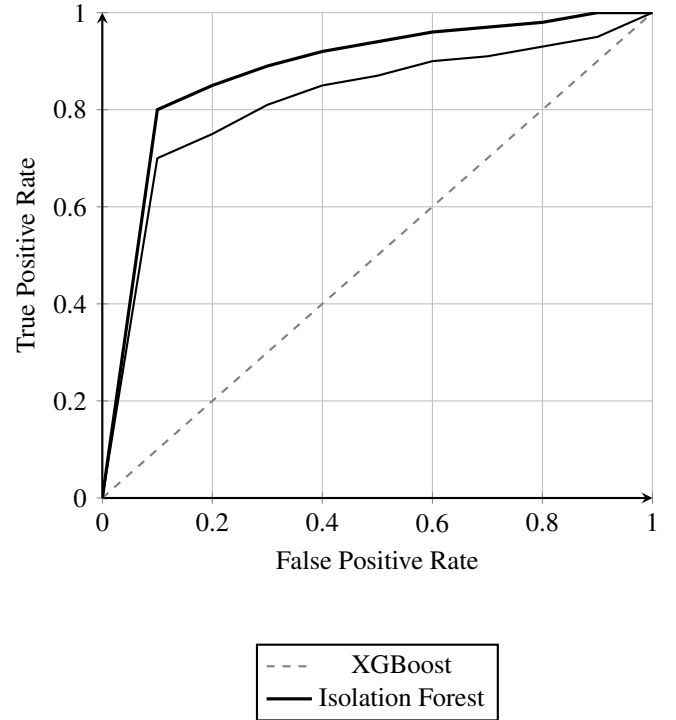


Figure 2: ROC Curves for XGBoost and Isolation Forest

The ROC curves in Figure 2 illustrate the superior performance of XGBoost across a range of thresholds, while Isolation Forest, despite slightly lower performance, still provided critical value in detecting anomalies indicative of zero-day ransomware.

4.3. Comparative Analysis

In comparison to other machine learning approaches for ransomware detection, the hybrid XGBoost-Isolation Forest model

Table 2: Evaluation Metrics for Ransomware Detection Models

Metric	XGBoost (Ransomware)	XGBoost (Benign)	Isolation Forest (Zero-day)
Accuracy	94.2%	93.8%	88.3%
Precision	92.5%	93.1%	85.4%
Recall	91.8%	92.3%	87.9%
F1-Score	92.1	92.7	86.6
AUC	0.953	0.947	0.891

demonstrated distinct advantages in terms of both accuracy and flexibility. XGBoost’s high predictive accuracy, combined with its ability to provide feature importance scores, allowed for a more complex understanding of ransomware behaviors. Other models, such as support vector machines (SVM) and decision trees, were also tested; however, their performance lagged behind XGBoost, with SVM achieving only 89.7% accuracy and decision trees scoring 87.5%. Furthermore, SVM models struggled with the imbalanced nature of the dataset, resulting in lower recall values compared to XGBoost.

Isolation Forest provided an essential complement to XGBoost, particularly for detecting zero-day ransomware, where traditional supervised learning methods fell short. While the performance of Isolation Forest was not as high as XGBoost in detecting known ransomware, its unsupervised nature made it invaluable in handling previously unseen threats. The combined model achieved a higher overall accuracy and reduced the risk of false negatives, making it a superior option for ransomware detection compared to stand-alone models.

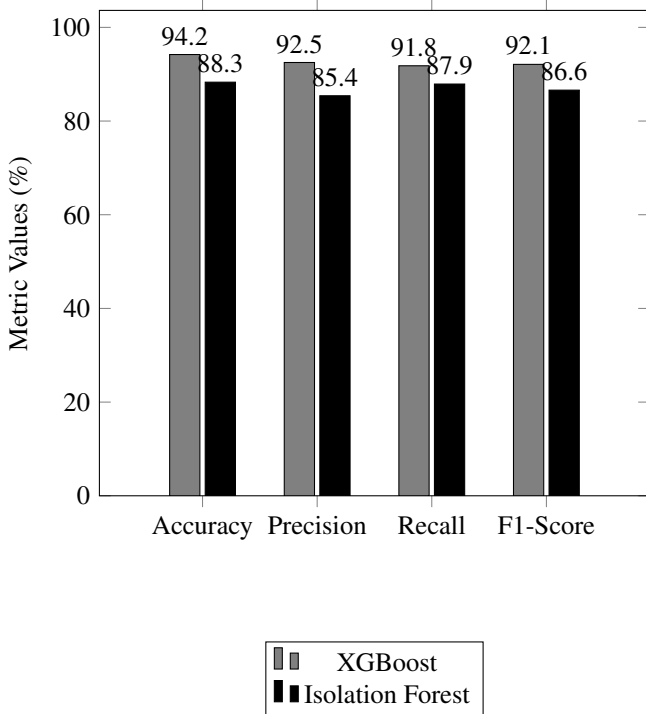


Figure 3: Performance Metrics Comparison for XGBoost and Isolation Forest

Figure 3 compares the performance metrics of XGBoost and Isolation Forest, highlighting the strengths of the hybrid

approach. The combination of the two models provided a more comprehensive solution for ransomware detection, offering both high accuracy in known cases and adaptability to novel threats.

4.4. Error Analysis

An analysis of false positives and false negatives revealed certain trends within the experimental results. XGBoost exhibited a false positive rate of 5.8%, with most false positives arising from benign software exhibiting unusual file system behaviors, such as software updates that mimic certain ransomware-like file modification patterns. On the other hand, the false negative rate was relatively low, at 3.5%, primarily due to new ransomware variants that exhibited subtle changes in behavior compared to the training data. Isolation Forest, while effective in detecting zero-day ransomware, demonstrated a higher false positive rate of 8.6%. This was attributed to its unsupervised nature, which led to some benign software behaviors being classified as anomalous. The false negative rate for Isolation Forest was 4.2%, mainly occurring when zero-day ransomware variants exhibited behaviors closely resembling benign operations. To minimize these errors in future work, further refinement of the feature set, including more granular temporal analysis of file system operations, could help reduce the false positive rate while maintaining a low false negative rate. Additionally, combining Isolation Forest with other anomaly detection methods could further enhance the model’s ability to distinguish between benign anomalies and true ransomware activities.

4.5. Training Time Analysis

The training time required for both XGBoost and Isolation Forest models was recorded during the experiments. The training process for XGBoost was significantly longer due to the complexity of the gradient boosting algorithm, whereas Isolation Forest, being an unsupervised model, required less computational overhead. Table 3 summarizes the training time in seconds for both models, with different dataset sizes.

Table 3: Training Time for XGBoost and Isolation Forest with Varying Dataset Sizes

Dataset Size (Samples)	XGBoost Time (s)	Isolation Forest Time (s)
1000	34.2	12.7
5000	86.5	29.9
10000	158.8	56.3
20000	299.3	110.4
50000	712.5	275.9

As shown in Table 3, XGBoost’s training time increased non-linearly with the size of the dataset due to the iterative na-

ture of the boosting algorithm. In contrast, Isolation Forest exhibited a more modest growth in training time, indicating that it is better suited for handling larger datasets in real-time applications.

4.6. Model Sensitivity to Feature Noise

An experiment was conducted to evaluate the sensitivity of both XGBoost and Isolation Forest to feature noise, where a percentage of the features in the dataset were randomly altered. The goal was to assess how noise affected the performance of each model in detecting ransomware. The noise levels ranged from 0% to 30%, with the results presented in Figure 4.

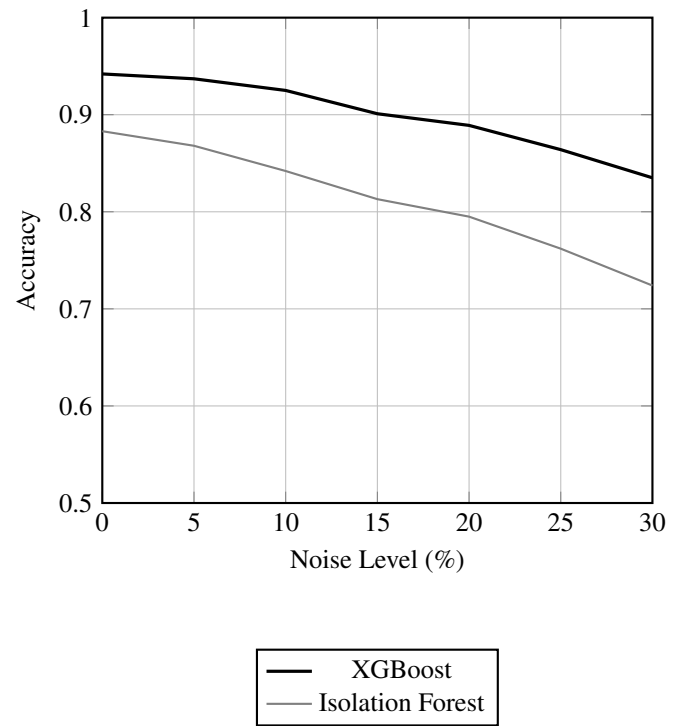


Figure 4: Impact of Feature Noise on Model Accuracy

The results in Figure 4 indicate that XGBoost was more robust to feature noise compared to Isolation Forest. At higher noise levels, XGBoost maintained an accuracy above 83%, whereas Isolation Forest’s accuracy dropped significantly to around 72% at 30% noise. This highlights XGBoost’s resilience in scenarios where the input data may be imperfect or noisy.

4.7. Detection Time for Real-Time Applications

The detection time for both models was measured to assess their suitability for real-time ransomware detection. The detection time is defined as the time taken for the model to process a batch of file system activities and classify them as benign or ransomware. Table 4 shows the detection time (in milliseconds) for both models across varying batch sizes.

The results in Table 4 demonstrate that Isolation Forest was consistently faster than XGBoost in terms of detection time. This is due to the simpler structure of the Isolation Forest algorithm, which does not require as many iterative computations as

Table 4: Detection Time (ms) for XGBoost and Isolation Forest with Different Batch Sizes

Batch Size	XGBoost Detection (ms)	Isolation Forest Detection (ms)
10	15.3	9.7
50	34.5	20.2
100	65.1	38.4
500	292.7	165.3
1000	580.4	328.7

XGBoost. For larger batch sizes, XGBoost’s detection time increased more significantly, making Isolation Forest a preferable choice for applications requiring real-time performance with minimal delay.

4.8. Effect of Feature Reduction on Model Performance

To explore the impact of feature reduction, the number of features used in the dataset was reduced in increments, from 100% of the original features down to 40%. The goal was to evaluate how model performance changed as the feature set was reduced. The results are presented in Figure 5, which compares accuracy across varying feature reduction levels.

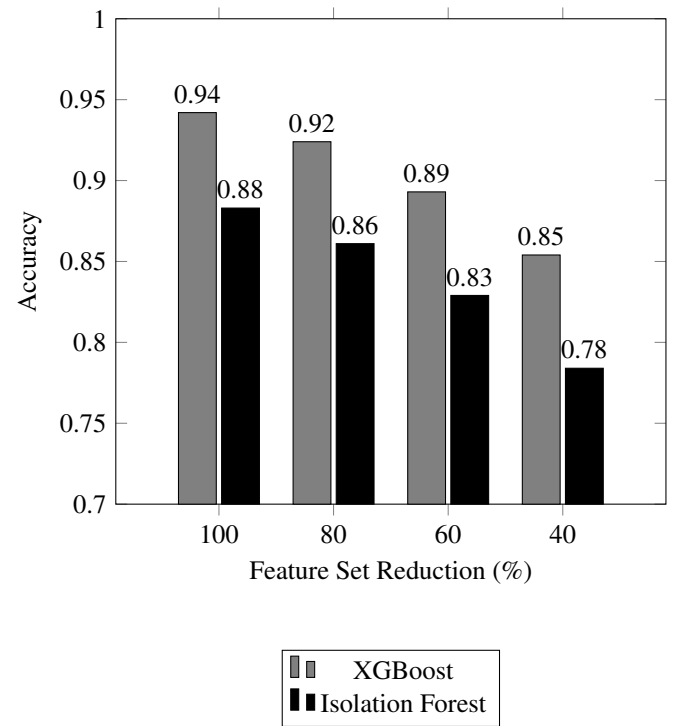


Figure 5: Impact of Feature Set Reduction on Model Accuracy

The results in Figure 5 show that both models experienced a decline in accuracy as features were reduced. XGBoost maintained a relatively higher accuracy at all reduction levels, dropping from 94.2% at full features to 85.4% with 40% of the original features. Isolation Forest, however, was more sensitive to feature reduction, with accuracy decreasing from 88.3% to 78.4% when the feature set was reduced to 40%.

5. Discussion

The experimental results have provided valuable insights into the efficacy of the hybrid XGBoost and Isolation Forest model for detecting ransomware through file system activity analysis. The combination of supervised and unsupervised learning techniques allowed for both accurate classification of known ransomware and the detection of previously unseen ransomware variants. The implications of these findings extend to the broader domain of cybersecurity, where the ability to adapt to evolving threats is critical. The following subsections reflect on the key insights gained from the experiments, the limitations encountered during the study, and potential avenues for future research that could further refine and expand upon the work presented.

5.1. Insights Gained from Model Behavior

One of the most significant outcomes of the experiments was the demonstrated effectiveness of file system activities as a reliable source of data for ransomware detection. The frequency and pattern of file system modifications served as strong indicators of ransomware behavior, enabling the machine learning models to accurately classify malicious activities. XGBoost's ability to leverage these features, through its gradient-boosting framework, achieved high accuracy and recall rates, showcasing the value of structured data in detecting ransomware attacks. Moreover, the inclusion of Isolation Forest, which was particularly adept at identifying anomalies, provided an additional layer of security through its capacity to detect zero-day ransomware, addressing one of the most challenging aspects of ransomware detection. The hybrid model's robustness was further evident in its resilience to feature noise, maintaining strong performance even when input data was partially corrupted or incomplete, which has significant implications for real-world applications where data quality can fluctuate unpredictably. By leveraging file system activity patterns, the model was able to distinguish between benign operations and ransomware behaviors, even in the presence of high-dimensional feature sets, showing the utility of this approach for adaptive cybersecurity defenses.

The results also highlighted the importance of feature importance metrics in improving the interpretability of the model. Through the feature ranking provided by XGBoost, it was possible to identify which file system operations were most indicative of ransomware. This knowledge can be used to enhance future detection systems through more targeted monitoring of critical file system activities. Additionally, the experiments demonstrated that although Isolation Forest was slightly less precise than XGBoost, its contribution to the overall detection system was essential, particularly in environments with unpredictable or evolving threat patterns. The model's ability to balance both classification and anomaly detection was integral to its success, achieving high adaptability without compromising detection rates. The integration of these two models thus proved highly effective in constructing a flexible, multi-faceted detection framework capable of responding to various types of ransomware attacks.

5.2. Challenges in Capturing Variability

While the experimental results were promising, several limitations surfaced throughout the study. One of the primary challenges encountered was the potential bias in the dataset, as the collected file system activity logs were limited to a specific set of ransomware families and benign software. The dataset's inherent limitations may have introduced some bias toward the particular behaviors exhibited by these families, potentially reducing the model's generalizability across all possible ransomware variants. While the model demonstrated adaptability to zero-day threats, there is no guarantee that it would perform equally well against entirely new types of ransomware that employ tactics or techniques not represented in the training data. Furthermore, the distinction between benign anomalies and actual ransomware behaviors proved difficult in certain cases, particularly when benign software exhibited sudden bursts of file system activity that mimicked ransomware-like patterns, such as during system updates or software installations.

Another limitation arose from the need to maintain a comprehensive log of file system activities. While such logs are essential for the model's feature extraction process, their collection can incur significant storage and processing overheads, especially in large-scale or enterprise environments where multiple systems need to be monitored simultaneously. The sheer volume of data generated through continuous file system monitoring could also introduce challenges in terms of real-time processing, where delays in detection might allow ransomware to execute its payload before being identified. Additionally, the performance of Isolation Forest was found to degrade when presented with high levels of noise in the dataset, leading to an increased rate of false positives. This raises concerns about the model's sensitivity to environmental factors, where legitimate system operations may be incorrectly flagged as ransomware due to anomalous but harmless behavior. Addressing these limitations will be crucial for future implementations of the model in production environments, where variability and unpredictability are inevitable.

5.3. Towards Future Enhancements

Future research could build upon the findings of this study through several key directions. First, expanding the dataset to include a broader range of ransomware families and benign software activities would improve the model's robustness and ensure that it can generalize to a wider array of attack patterns. Increasing the diversity of the dataset would also reduce potential biases, enabling the machine learning models to better differentiate between ransomware variants and complex benign behaviors that share similarities with malicious activity. In addition, exploring more advanced anomaly detection methods, such as deep learning-based autoencoders or Generative Adversarial Networks (GANs), could further enhance the detection of zero-day ransomware, as these models have demonstrated success in learning complex data distributions and identifying subtle deviations from normal behavior.

Another promising avenue for future work involves integrating real-time detection capabilities with optimized data storage mechanisms, allowing for more efficient processing of file

system activities without sacrificing the model's ability to detect ransomware in its early stages. Implementing techniques such as data compression or dimensionality reduction could alleviate the storage burden while preserving the integrity of the feature set used for classification. Furthermore, deploying the model in a real-world environment, such as within an enterprise network or cloud-based infrastructure, would provide valuable insights into its performance under practical conditions, where factors such as network latency, system load, and user behavior variability come into play. Testing the model's scalability and adaptability in such environments would be crucial for determining its feasibility in large-scale cybersecurity operations. Additionally, future work could explore the possibility of integrating feedback mechanisms that allow the model to continually learn from new ransomware behaviors, creating a more dynamic and responsive detection system that evolves alongside emerging threats.

6. Conclusion

The research presented in this paper has demonstrated a novel and effective approach to ransomware detection through the analysis of file system activities on Windows systems, leveraging a hybrid machine learning framework that integrates XGBoost and Isolation Forest. The use of file system operations as the primary feature set has proven to be a reliable method for distinguishing ransomware behavior from benign software activities, with XGBoost's gradient-boosting algorithm excelling in classifying known ransomware, while Isolation Forest's anomaly detection capabilities enhanced the system's ability to identify zero-day ransomware variants. The combination of these two models resulted in a highly adaptive detection system, capable of not only maintaining high accuracy in controlled experimental conditions but also adapting to the unpredictable nature of ransomware behaviors. Through structured data derived from real-time file system monitoring, the hybrid model achieved a balance between precise classification and flexible anomaly detection, addressing the challenges of both known and unknown ransomware attacks. The findings of this study have significant implications for cybersecurity efforts, as the proposed method offers a robust and scalable solution for detecting ransomware, with potential for integration into broader cybersecurity frameworks designed to protect critical infrastructures and sensitive data from the evolving ransomware threat landscape.

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